LAPORAN PRAKTIKUM

MATA KULIAH:

PEMROGRAMAN MOBILE



Dosen Pengampu:

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Bagian D: Aplikasi "Hello Flutter"

```dart
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 1',
      home: Scaffold(
        appBar: AppBar(
          title: const Text('Hello Flutter'),
        ),
        body: const Center(
          child: Text(
            'Halo, nama s',
            style: TextStyle(fontSize: 24),
          ),
        ),
      ),
    );
  }
}
```

Hasil:



Bagian E: Widget Layout Dasar

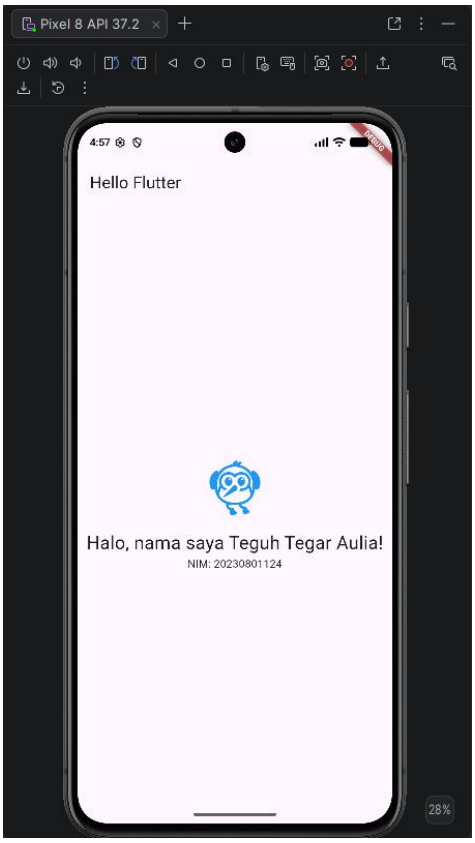
```dart
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 1',
      home: Scaffold(
        appBar: AppBar(
          title: const Text('Hello Flutter'),
        ),
        body: Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: const [
              Icon(
                Icons.flutter_dash,
                size: 80,
                color: Colors.blue,
              ),
              SizedBox(height: 16),
              Text(
                'Halo, nama saya Teguh Tegar Aulia!',
                style: TextStyle(fontSize: 24),
              ),
              Text('NIM: 20230801124'),
            ],
          ),
        ),
      ),
    );
  }
}
```

Hasil:



Bagian F: Widget Interaktif (StatefulWidget)

```dart
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 1',
      home: const CounterPage(),
    );
  }
}

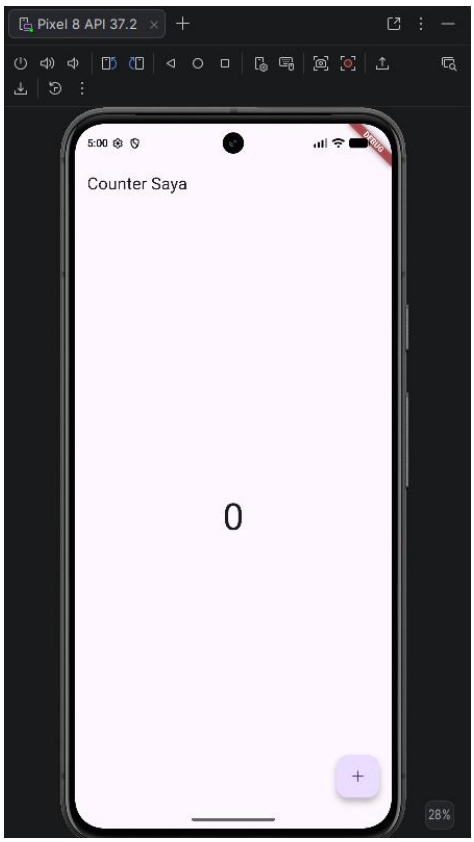
class CounterPage extends StatefulWidget {
  const CounterPage({super.key});

  @override
  State<CounterPage> createState() => _CounterPageState();
}

class _CounterPageState extends State<CounterPage> {
  int _count = 0;
```

```dart
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Counter Saya'),
      ),
      body: Center(
        child: Text(
          '$_count',
          style: const TextStyle(fontSize: 48),
        ),
      ),
      floatingActionButton: FloatingActionButton(
        onPressed: () => setState(() => _count++),
        child: const Icon(Icons.add),
      ),
    );
  }
}
```

Hasil:



Latihan Mandiri

1. Ubah warna AppBar dan warna teks sesuai selera (backgroundColor, color).
2. Tambahkan tombol kedua (Icons.remove) untuk mengurangi angka.
3. Tambahkan tombol reset yang mengembalikan angka ke 0.
4. Cegah angka menjadi negatif.

```dart
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'pertemuan_1',
      home: const CounterPage(),
    );
  }
}

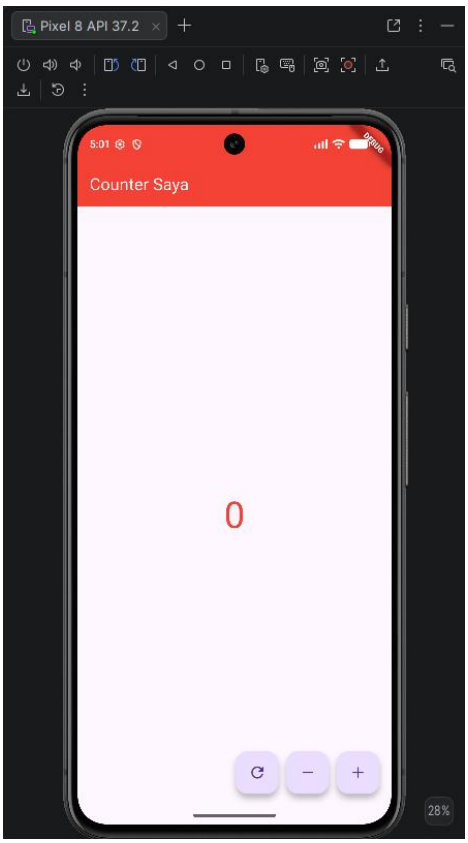
class CounterPage extends StatefulWidget {
  const CounterPage({super.key});

  @override
  State<CounterPage> createState() => _CounterPageState();
}

class _CounterPageState extends State<CounterPage> {
  int _count = 0;

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text(
          'Counter Saya',
          style: TextStyle(color: Colors.white),
        ),
        backgroundColor: Colors.red,
      ),
      body: Center(
        child: Text(
          '$_count',
          style: const TextStyle(
            fontSize: 48,
            color: Colors.red,
          ),
        ),
      ),
      floatingActionButton: Row(
        mainAxisSize: MainAxisSize.min,
        children: [
          FloatingActionButton(
            onPressed: () => setState(() => _count = 0),
            child: const Icon(Icons.refresh),
```

```
          ),
          const SizedBox(width: 10),
          FloatingActionButton(
            onPressed: () => setState(() {
              if (_count > 0) _count--;
            }),
            child: const Icon(Icons.remove),
          ),
          const SizedBox(width: 10),
          FloatingActionButton(
            onPressed: () => setState(() => _count++),
            child: const Icon(Icons.add),
          ),
        ],
      ),
    );
  }
}
```



Tugas Kartu Perkenalan

Buat aplikasi "Kartu Perkenalan" satu halaman yang menampilkan: foto/ikon, nama, NIM, jurusan, dan hobi menggunakan Column, Text, Icon, dan SizedBox. Kumpulkan berupa tangkapan layar aplikasi berjalan dan tautan repositori (atau berkas main.dart).

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
```

```dart
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Kartu Perkenalan',
      home: Scaffold(
        appBar: AppBar(
          title: const Text('Kartu Perkenalan'),
        ),
        body: Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: const [
              Icon(
                Icons.person,
                size: 100,
                color: Colors.blue,
              ),
              SizedBox(height: 20),
              Text(
                'Teguh Tegar Aulia',
                style: TextStyle(fontSize: 24),
              ),
              SizedBox(height: 10),
              Text('NIM: 20230801124'),
              SizedBox(height: 10),
              Text('Jurusan: Teknik Informatika'),
              SizedBox(height: 10),
              Text('Hobi: Bermain Game'),
            ],
          ),
        ),
      ),
    );
  }
}
```

Hasil:



Pertanyaan Refleksi:

1. Apa perbedaan StatelessWidget dan StatefulWidget?
2. Mengapa perubahan variabel _count perlu dibungkus setState()?
3. Apa keuntungan hot reload dibanding rebuild penuh?

Jawaban:

1. StatelessWidget itu widget yang tampilannya tidak berubah berdasarkan data yang ada di dalamnya, sedangkan StatefulWidget bisa berubah ketika datanya berubah. contohnya CounterPage karena angka _count bisa bertambah.
2. _count perlu dibungkus setState() supaya flutter tahu bahwa datanya berubah dan menampilkan ulang tampilan dengan nilai terbaru.
3. Hot reload lebih cepat karena kita bisa li perubahan kode langsung di aplikasi yang sedang berjalan tanpa harus menjalankan ulang aplikasi dari awal.